

CPEN 416

Lecture 08: The oracle, query complexity, and Deutsch's algorithm

Thursday 8 October 2026

Announcements

★ Scott Aaronson Lecture
today @ 3:30 KACS 2020/2030

- Midterm grading in progress
- Project details coming soon (start thinking about groups)
- No tutorial Monday (holiday)
- Quiz 4 on Tuesday

↓
Today's content

↓
5 groups 4
2 groups 3
(use Piazza to find group)

Module 2 learning outcomes

Learning outcomes:

- explain what it means for an algorithm to have a quantum speedup
- define quantum oracles and query complexity
- implement oracles and Grover's algorithm in PennyLane
- identify the different components of the quantum compilation stack
- define and list common universal gate sets
- estimate the resources required to run a quantum algorithm
- perform simple circuit optimizations in PennyLane

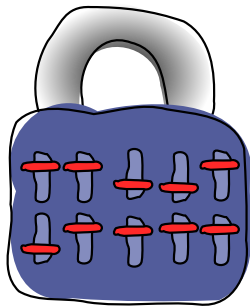
Today

Learning outcomes:

- Define the query complexity of an algorithm
- Describe multiple strategies for implementing an *oracle* query with a quantum circuit
- Apply the phase kickback trick
- Implement Deutsch's algorithm to determine whether a Boolean function is constant or balanced (in theory and in PennyLane)

Oracles: motivating problem

Suppose we would like to find the combination for a “binary” lock:



How do we solve this classically? 2^n tries (worst case)

Image credit: Codebook node A.1

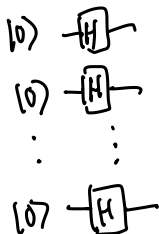
Idea: use superposition

Can we do better with a quantum computer?

Idea: take n qubits and construct a *uniform* superposition of all combinations

$$\frac{1}{\sqrt{2}^n} (|0\dots 0\rangle + |0\dots 01\rangle + |00\dots 010\rangle + \dots + |11\dots 1\rangle)$$

Exercise: design a circuit to create this state.

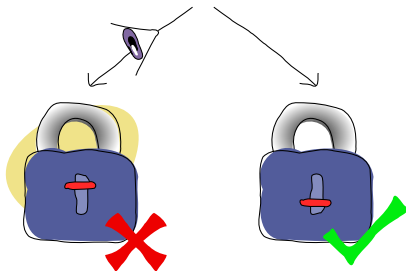


$\rightarrow$

$$\begin{aligned} n=2 \quad & \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \\ & = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle) \end{aligned}$$

Idea: use superposition

Measurements are probabilistic - the solution just being “in” the superposition doesn’t make it easier to solve the problem.



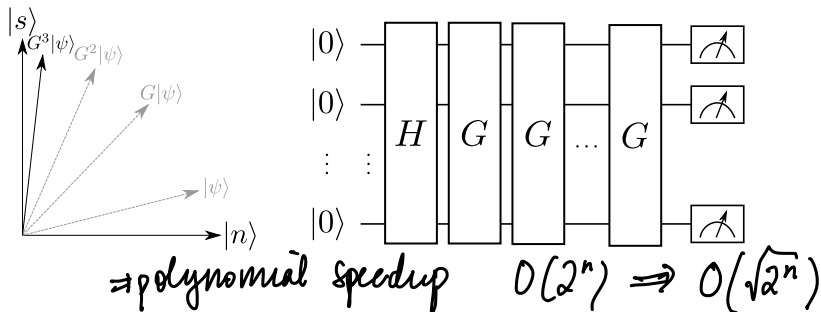
Quantum computers are **NOT** faster because they “compute everything at the same time.”

Image credit: Codebook node A.1

Solving problems with quantum computers

Can we solve this problem better with a quantum computer?

Yes: **amplitude amplification**, and **Grover's algorithm**



We will explore the algorithmic primitives, and see other cases where quantum computers have a **speed up in query complexity**.

Oracles

Revisit the lock problem and model a “try” as a function. Let

- $\mathbf{x}$ be an n -bit string (an input combination)
- $\mathbf{s}$ be the correct combination

$$f(\vec{x}) = \begin{cases} 1 & \vec{x} = \vec{s} \\ 0 & \text{otherwise.} \end{cases}$$

$\vec{x} \in \{0,1\}^n$ n bit string

Don't care *how* $f(\mathbf{x})$ is evaluated, only that it gives a “yes/no” answer. $f(\mathbf{x})$ is a black box, or **oracle**.

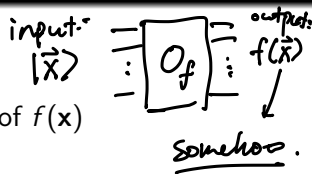
Every time we try a combination, we are **querying the oracle**.
The amount of queries is the **query complexity**.

classical query complexity: $O(2^n)$

Quantum oracles

Must determine

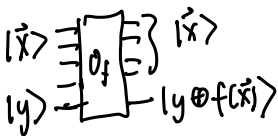
- a mathematical description of an oracle
- where to record the output, i.e., the value of $f(x)$



Idea 1: encode the result in the state of an additional qubit.

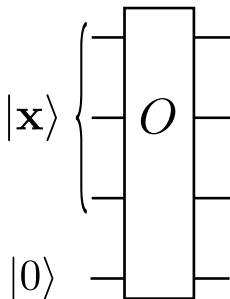
$$O_f \left(\underbrace{|\vec{x}\rangle}_{n \text{ qubits}} \underbrace{|y\rangle}_{1 \text{ qubit}} \right) = |\vec{x}\rangle |y \oplus f(\vec{x})\rangle$$

Exercise: Consider a 2-qubit system where $f(01) = 1$, and $f(x) = 0$ for all other x . What is the action of the oracle?



Input state	Output state
$ 00\rangle 0\rangle$	$ 00\rangle 0\rangle$
$ 01\rangle 0\rangle$	$ 01\rangle 1\rangle$
$ 10\rangle 0\rangle$	$ 10\rangle 0\rangle$
$ 11\rangle 0\rangle$	$ 11\rangle 0\rangle$

Quantum oracles



$$O|000\rangle|0\rangle = |000\rangle|0\rangle$$

$$O|001\rangle|0\rangle = |001\rangle|0\rangle$$

$$O|010\rangle|0\rangle = |010\rangle|0\rangle$$

$$O|011\rangle|0\rangle = |011\rangle|0\rangle$$

$$O|100\rangle|0\rangle = |100\rangle|0\rangle$$

$$O|101\rangle|0\rangle = |101\rangle|0\rangle$$

$$O|110\rangle|0\rangle = |110\rangle|1\rangle$$

$$O|111\rangle|0\rangle = |111\rangle|0\rangle$$

Quantum oracles

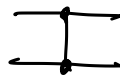
Idea 2: encode the result in the phase

$$O_f |\vec{x}\rangle = (-1)^{f(\vec{x})} |\vec{x}\rangle$$

$$|\vec{x}\rangle \equiv \boxed{O_f} (-1)^{f(\vec{x})} |\vec{x}\rangle$$

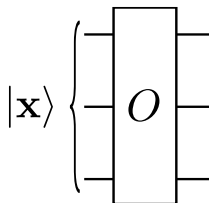
Exercise: Consider a 2-qubit system where $f(11) = 1$, and $f(\mathbf{x}) = 0$ for all other $\mathbf{x}$. What is the action of the oracle?

Input state	Output state
$ 00\rangle$	$ 00\rangle$
$ 01\rangle$	$ 01\rangle$
$ 10\rangle$	$ 10\rangle$
$ 11\rangle$	$- 11\rangle$



CZ gate!

Quantum oracles



$$O|000\rangle = |000\rangle$$

$$O|001\rangle = |001\rangle$$

$$O|010\rangle = |010\rangle$$

$$O|011\rangle = |011\rangle$$

$$O|100\rangle = |100\rangle$$

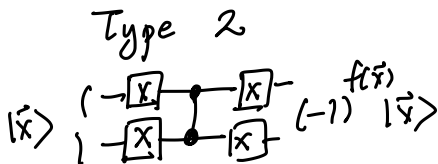
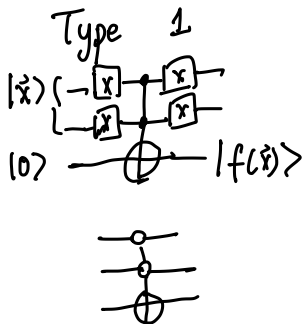
$$O|101\rangle = |101\rangle$$

$$O|110\rangle = -|110\rangle$$

$$O|111\rangle = |111\rangle$$

Oracle circuits

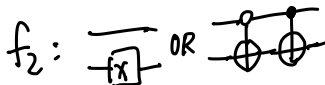
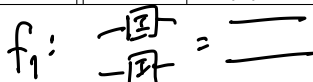
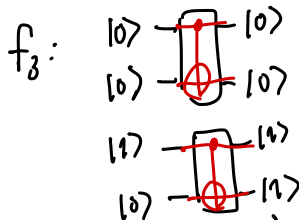
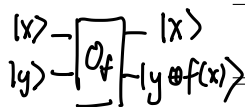
Exercise: develop two quantum circuit implementations for a 2-input oracle with function $f(00) = 1$.



Oracle circuits

Exercise: construct two-qubit oracle circuits (“type 1” oracle) for the following four functions.

Name	Action	Name	Action
f_1	$f_1(0) = 0$ $f_1(1) = 0$	f_2	$f_2(0) = 1$ $f_2(1) = 1$
f_3	$f_3(0) = 0$ $f_3(1) = 1$	f_4	$f_4(0) = 1$ $f_4(1) = 0$



$$\vee 0 \oplus f(1) = 0 \oplus 1 = 1$$

Deutsch's algorithm

Motivation: You have access to an oracle that implements one of the following functions:

Name	Action	Name	Action	
f_1	$f_1(0) = 0$ $f_1(1) = 0$	f_2	$f_2(0) = 1$ $f_2(1) = 1$	} <i>constant</i>
f_3	$f_3(0) = 0$ $f_3(1) = 1$	f_4	$f_4(0) = 1$ $f_4(1) = 0$	

Functions f_1 and f_2 are *constant*, f_3 and f_4 are *balanced*.

How many **classical** oracle queries to determine if the oracle function is constant or balanced? How many **quantum** queries?

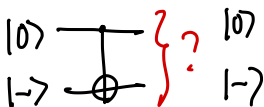
classical : 2 quantum : 1

Phase kickback

The secret is the *phase kickback* trick.

Exercise: apply CNOT_{01} to the following two-qubit states.

$$|0\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right), \quad |1\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$



$$|1\rangle |-\rangle = \frac{1}{\sqrt{2}} (|10\rangle - |11\rangle) \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}} (|11\rangle - |10\rangle) = |1\rangle \left(-\frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \right) = -|1\rangle |-\rangle$$

$$|0\rangle |-\rangle \rightarrow |0\rangle |-\rangle$$

$$|1\rangle |-\rangle \rightarrow -|1\rangle |-\rangle$$

$$|b\rangle |-\rangle \rightarrow (-1)^b |b\rangle |-\rangle$$

phase
kickback

Deutsch's algorithm

Let U_f be an oracle for one of the four functions:

$$U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$$

Initializing the second qubit to $|-\rangle$ allows us to learn $f(0) \oplus f(1)$ with a single oracle query.

Q: How does $f(0) \oplus f(1)$ relate to properties of the function?

balanced: $f(0) \oplus f(1) = 1$

constant: $f(0) \oplus f(1) = 0$

Deutsch's algorithm

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$$

$$\begin{aligned} U_f |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) &= U_f \left(\frac{1}{\sqrt{2}} |x\rangle |0\rangle - \frac{1}{\sqrt{2}} |x\rangle |1\rangle \right) \\ &= \frac{1}{\sqrt{2}} \left(|x\rangle |0 \oplus f(x)\rangle - |x\rangle |1 \oplus f(x)\rangle \right) \end{aligned}$$

If $f(x) = 0$,

$$U_f |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = |x\rangle |-\rangle$$

If $f(x) = 1$,

$$U_f |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = - |x\rangle |-\rangle$$

Deutsch's algorithm

Remember how we generalized the result for CNOT:

$$\text{CNOT} \left(|b\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \right) = (-1)^b |b\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right),$$

So we can write

$$U_f \left(|x\rangle |-\rangle \right) = (-1)^{f(x)} |x\rangle |-\rangle$$

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$$

Deutsch's algorithm

How to use this to get $f(0) \oplus f(1)$?

$$\begin{aligned}
 & U_f \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\
 &= \frac{1}{\sqrt{2}} \left(U_f |0\rangle |-\rangle + U_f |1\rangle |-\rangle \right) \\
 &= \frac{1}{\sqrt{2}} \left((-1)^{f(0)} |0\rangle |-\rangle + (-1)^{f(1)} |1\rangle |-\rangle \right) \\
 &= \frac{1}{\sqrt{2}} \left((-1)^{f(0)} |0\rangle + (-1)^{f(1)} |1\rangle \right) |-\rangle \\
 &= \frac{1}{\sqrt{2}} (-1)^{f(0)} \left(|0\rangle + (-1)^{f(1) \oplus f(0)} |1\rangle \right) |-\rangle
 \end{aligned}$$

Deutsch's algorithm

$$U_f \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \frac{|0\rangle + (-1)^{f(0) \oplus f(1)} |1\rangle}{\sqrt{2}} \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

If the function is constant, $f(0) \oplus f(1) = 0$ and the state is

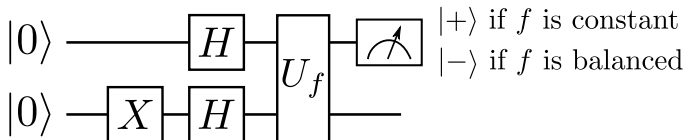
$$U_f \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = |+\rangle |-\rangle$$

But if the function is balanced, $f(0) \oplus f(1) = 1$ and the state is

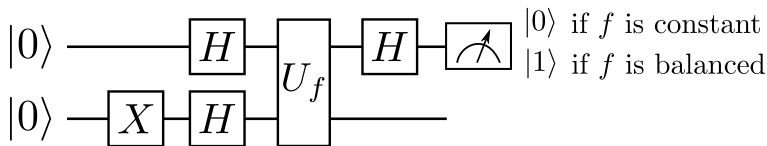
$$U_f \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = |-\rangle |-\rangle$$

Deutsch's algorithm

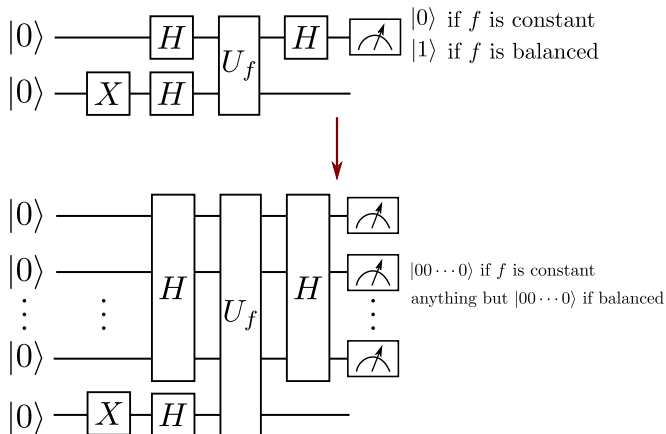
Deutsch's algorithm as a circuit:



Equivalently,



Generalization: Deutsch-Jozsa algorithm



$2^{n-1} + 1$ classical queries in worst case; still only 1 quantum query.
(Challenge: try implementing it yourself to check if this works!)

Oracle-based algorithms

A few other interesting algorithms:

Bernstein-Vazirani algorithm (practice A2)

Given $f : \{0, 1\}^n \rightarrow \{0, 1\}$ such that $f(x) = x \cdot s$ for some secret bitstring s . Find s using the fewest number of oracle queries.

Simon's algorithm

Given $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$ and promised that for some non-trivial bit string s , $f(x) = f(y)$ iff $x \oplus y = s$. Find s using the fewest number of oracle queries.

Next time

Content:

- Quiz 4 on today's material
- Amplitude amplification and Grover's algorithm

Action items:

- ① Start thinking about project groups

Recommended reading:

- For today: Codebook module BA; Nielsen & Chuang 1.4.1-1.4.4
- For next class: Codebook module GA; Nielsen & Chuang 6.1